

LECTURE 7: MATH1061/7861
TOPICS: MULTIPLE QUANTIFIERS
DATE: TUESDAY MARCH 10, 2026

Learning objectives:

- (1) Understand and negate statements with multiple quantifiers.

$\forall, \exists$

* Applied Class Exercise sheets

↳ If you don't know how to start / proceed, ask other students or the tutor

* If your class falls on public holiday, you must go to a different ~~to~~ Applied class in the same week.

Q1:

(1) True or false:

y depends on x.

$$\forall x \in \mathbb{Z}, \exists y \in \mathbb{Z} \text{ such that } y = 2x.$$

True.

Because twice an integer is an integer.

(2) Now we change the order. True or False:

y cannot depend on x.

$$\exists y \in \mathbb{Z} \text{ such that } \forall x \in \mathbb{Z}, y = 2x.$$

False.

For instance, take

$$x = 1$$

then we must have $y = 2$.

Next, take $x = 2$ then we must have $y = 4$.

So there is no single y that does the job for all x .

(3) Now we change the roles of x and y . True or False:

$$\forall x \in \mathbb{Z}, \exists y \in \mathbb{Z} \text{ such that } x = 2y$$

False.

Take $x=1$ then y must
be $\frac{1}{2}$. But $\frac{1}{2} \notin \mathbb{Z}$.

Q4: Write the negation, and determine whether the original statement is true or false.

y depends on x .

$\forall x \in \mathbb{R}, \exists y \in \mathbb{R}$ such that $x + y = 0$

Negation $\exists x \in \mathbb{R}, \forall y \in \mathbb{R}, x + y \neq 0$.

Original statement is True.

~~Be~~ set Because we can set

$y = -x$ and negative of real numbers is real.

$\exists !$ $\leftarrow$ There exists unique.

$\forall x \in \mathbb{Z}, \exists y \in \mathbb{Z}, y > x$ True

$\forall x \in \mathbb{Z}, \exists ! y \in \mathbb{Z}, y > x$ False.

Negation: $\exists x \in \mathbb{Z}, \dots$

Think about this.

Q5: Write the negation, and determine whether the original statement is true or false.

$$\forall x \in \mathbb{R}, \exists y \in \mathbb{R} \text{ such that } xy \neq 0$$

False. If we take $x=0$, then there is no y for which $xy \neq 0$.

Negation: $\exists x \in \mathbb{R}, \forall y \in \mathbb{R}, xy = 0$.